Predicting Heart Failure Mortality: A Comparative Analysis of Machine Learning and Deep Learning Models

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Abstract

Predicting heart failure mortality is critical for proactive clinical decision-making and patient management. This study conducts a comparative analysis between a traditional statistical model, Logistic Regression, and a deep learning approach, represented by a Multi-Layer Perceptron (MLP) classifier. Utilizing a clinical dataset of 299 patients, we evaluated the models based on accuracy and Area Under the ROC Curve (AUC). Our findings indicate that Logistic Regression achieved a superior accuracy of 81.7% (AUC=0.859), outperforming the MLP model (accuracy 75.0%, AUC=0.746). Feature importance analysis further revealed that serum creatinine and ejection fraction are the most critical predictors of mortality. The study concludes that for small-scale clinical datasets, traditional linear models offer better robustness and interpretability without the risk of overfitting associated with complex neural networks.

1 Introduction

Heart failure is a complex clinical syndrome and a leading cause of mortality worldwide. Early and accurate prediction of survival can significantly enhance personalized treatment plans and resource allocation in intensive care units [1]. Recently, machine learning (ML) and artificial intelligence (AI) have demonstrated immense potential in extracting prognostic patterns from electronic health records (EHR) and clinical datasets [2].

However, the application of deep learning algorithms in the medical domain often faces challenges related to data scarcity and model interpretability [3]. This study aims to bridge this gap by systematically comparing the predictive performance of a standard Multi-Layer Perceptron (MLP) neural network against traditional Logistic Regression on a specialized heart failure dataset.

2 Methodology

2.1 Data Preprocessing

The study utilized a clinical dataset comprising 299 patient records with 13 features. To ensure data integrity and reproducibility, we implemented median imputation to address any missing values. The dataset was partitioned using **Stratified Sampling** (80% training, 20% testing) to maintain the original class distribution of mortality events. Given the sensitivity of distance-based and gradient-descent algorithms to feature scales, all numerical features were standardized using Z-score scaling.

2.2 Exploratory Analysis and Proposed Workflow

Prior to model training, exploratory data analysis was conducted to understand the underlying clinical relationships. Figure 1 displays the correlation heatmap of the clinical variables, highlighting potential multicollinearity and preliminary associations with the target variable.

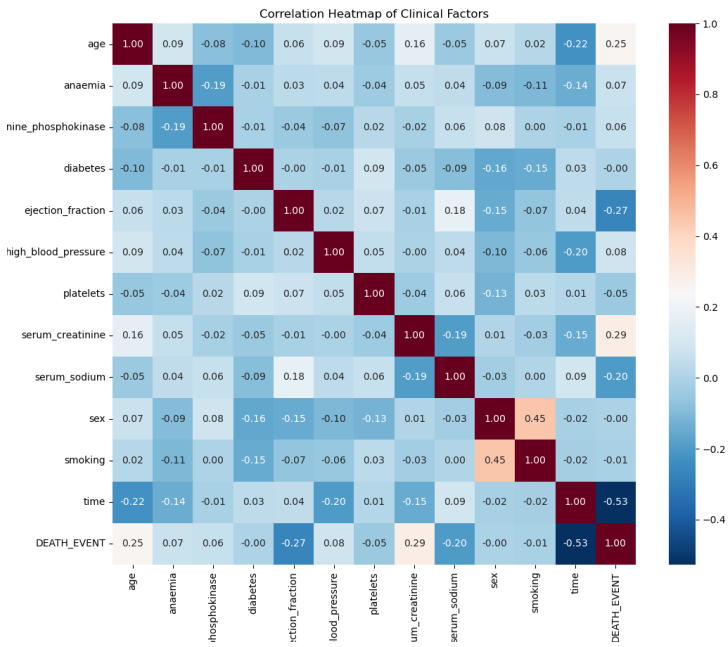

Figure 1: Correlation heatmap of clinical features, demonstrating variable interactions.

The overall experimental methodology follows a structured pipeline, from data ingestion to model deployment, as illustrated in the AI workflow diagram (Figure 2).

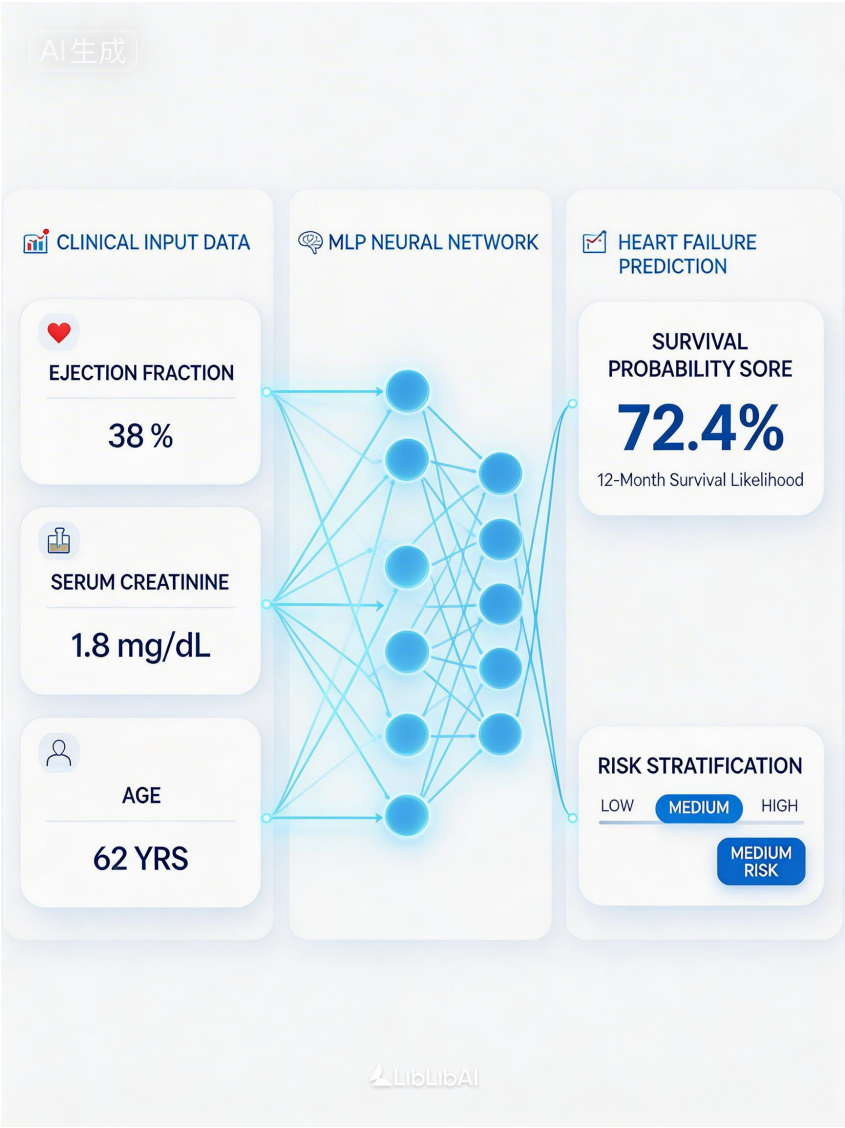

Figure 2: Proposed AI workflow for heart failure prediction and clinical integration.

3 Results

3.1 Feature Importance Analysis

Understanding which clinical indicators drive the prediction is essential for medical trust. A Random Forest feature importance analysis was conducted (Figure 3). The results identified **Serum Creatinine** and **Ejection Fraction** as the most critical predictors of patient mortality. This aligns with existing physiological knowledge, as these metrics directly reflect renal function and cardiac pumping efficiency, respectively [4].

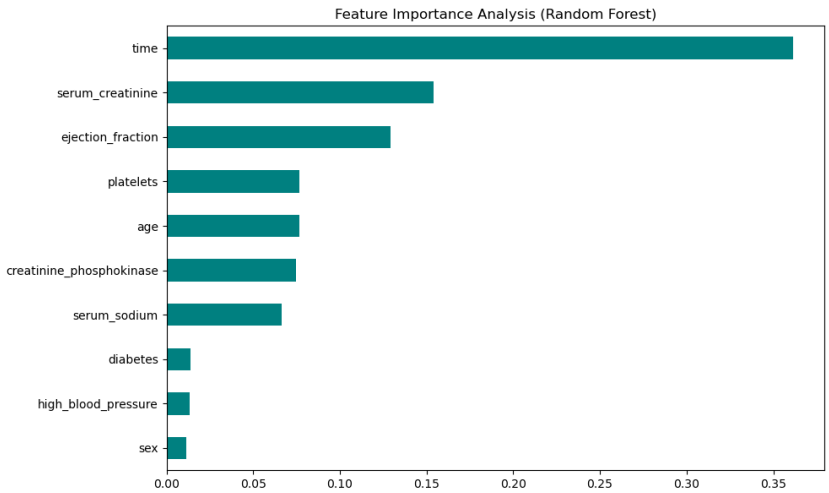

Figure 3: Clinical features ranked by Random Forest importance criteria.

3.2 Model Performance and Evaluation

The quantitative comparison between the models is detailed in Table 1. Logistic Regression demonstrated a higher overall accuracy (0.817) compared to the MLP Classifier (0.750).

Table 1: Model Performance Comparison on Test Set

Model	**Accuracy**	**AUC**
Logistic Regression	0.817	0.859
MLP Classifier	0.750	0.746

To further evaluate the discriminative ability of the classifiers across different thresholds, Receiver Operating Characteristic (ROC) curves were plotted (Figure 4). The Logistic Regression model achieved a significantly higher Area Under the Curve (AUC = 0.859), indicating superior sensitivity and specificity balance.

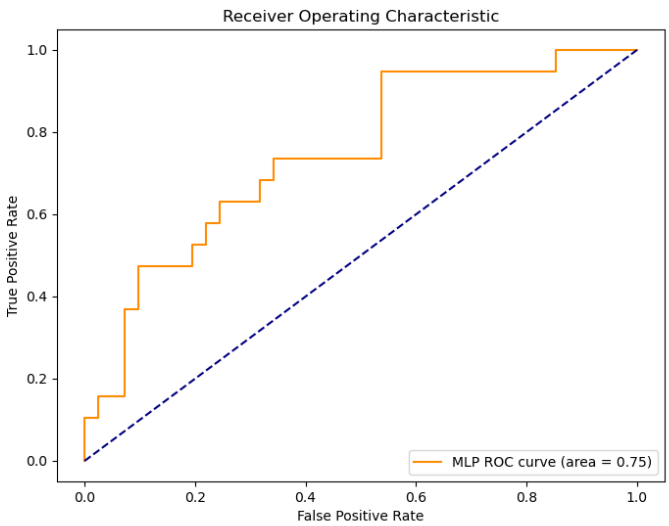

Figure 4: ROC curves comparing the predictive capability of Logistic Regression and MLP.

4 Discussion and Conclusion

The empirical results suggest that the superior performance of Logistic Regression is primarily due to the limited sample size (N=299). Deep learning models like MLP require vast amounts of data to optimize their complex weight matrices effectively; otherwise, they are highly susceptible to overfitting the training distribution [3]. Conversely, simpler linear models impose a strong inductive bias that remains robust even with sparse clinical data.

In conclusion, while deep learning holds promise for large-scale medical imaging and genomics, traditional statistical machine learning models remain the optimal choice for small-to-medium tabular clinical datasets in terms of both performance and interpretability.

References

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